STATISTICS PROJECT REPORT

From Data to Insights to Reporting

Probability, Normal Distribution, Hypothesis Testing and ANOVA

Introduction

This report covers the four problems I worked on in my statistics project. For every problem I followed the same four steps: first understand the business problem, then turn it into a data science (statistics) problem, then solve it in Python, and at the end turn the numbers back into a simple business answer. The code, outputs and graphs are in the attached HTML file (project_statistic.html).

Problem	Topic	Method used
1	Injuries among football players by playing position	Basic and conditional probability
2	Breaking strength of gunny bags	Normal distribution, Z-scores
3	Hardness of unpolished stones (Zingaro Company)	One sample t-test (left tailed)
4	Hardness of dental implants	Shapiro, Levene, one way and two way ANOVA, Tukey HSD

Tools: Python with pandas, numpy, scipy, statsmodels, seaborn and matplotlib. Significance level is 5% (0.05) everywhere.

Problem 1: Injuries Among Football Players

1. Understanding the Business Problem

A football club wants to know how injuries are spread across the positions of its players. The data has 235 players in four positions (Striker, Forward, Attacking Midfielder, Winger) and tells whether each player was injured or not. If the club knows which position gets hurt more often, the medical and training staff can plan better. The things to measure here are the number of injured players and the number of players in each position.

2. Translating it into a Data Science Problem

The data is a simple count table and it does not depend on time. We do not need a model here, we only want to understand the basic pattern, so the problem becomes finding four probabilities: the chance of injury, the chance of being a forward or a winger, the chance of being a striker and injured, and the chance of being a striker given the player is injured.

3. Solving the Data Science Problem

Predict before exploring

Before calculating I expected that more than half of the players would be injured, and that strikers and forwards would make up most of the injured group simply because they have the most players.

Analysis

Position	Injured	Not Injured	Total
Striker	45	32	77
Forward	56	38	94
Attacking Midfielder	24	11	35
Winger	20	9	29
Total	**145**	**90**	**235**

Question	Calculation	Answer
P(Injury)	145 / 235	0.6170 (61.7%)
P(Forward or Winger)	(94 + 29) / 235	0.5234 (52.34%)
P(Striker and Injured)	45 / 235	0.1915 (19.15%)
P(Striker | Injured)	45 / 145	0.3103 (31.03%)

Odd things and zooming in

The probability of a striker being injured and the probability of an injured player being a striker are not the same thing (19.15% against 31.03%). The first one divides by all 235 players, the second one divides only by the 145 injured players. It is easy to mix these two up, so I checked the denominator carefully each time.

4. Translating it back to a Business Solution

Around 6 out of 10 players in this data got injured, which is high. Strikers are almost one third (31%) of all injured players, and forwards and wingers together are more than half of the squad. The club should keep an eye on injury prevention for the attacking positions, especially strikers, and should keep the data updated so that injury rates per position (not only counts) can be compared in future.

Problem 2: Breaking Strength of Gunny Bags

1. Understanding the Business Problem

A company makes gunny bags and the quality depends on how much load a bag can take before it breaks. The breaking strength is measured in kg per sq cm. The company wants to know what share of its bags is too weak, and what share is in an acceptable range, so it can judge the quality of its production. The data given had two machines (Machine 1 and Machine 2) with 26 rows, but the questions were answered using the stated population values.

2. Translating it into a Data Science Problem

Breaking strength is a continuous variable and is assumed to follow a normal distribution with mean (mu) = 5 and standard deviation (sigma) = 1.5. Each question about a "proportion of bags" becomes an area under the normal curve, found by converting the value into a Z-score, Z = (x - mu) / sigma.

3. Solving the Data Science Problem

Predict before exploring

Since 5 is the mean, I expected about half of the bags below 5. I expected a small share (around 10%) below 3.17 because that value is more than one standard deviation under the mean, and a large share (around 80%) above 3.6.

Analysis

Question	Z-score	Proportion
Strength less than 3.17	-1.22	0.1112 (11.12%)
Strength at least 3.6	-0.933	0.8247 (82.47%)
Strength between 5 and 5.5	0 and 0.333	0.1306 (13.06%)
Strength NOT between 3 and 7.5	-1.333 and 1.667	0.1393 (13.93%)

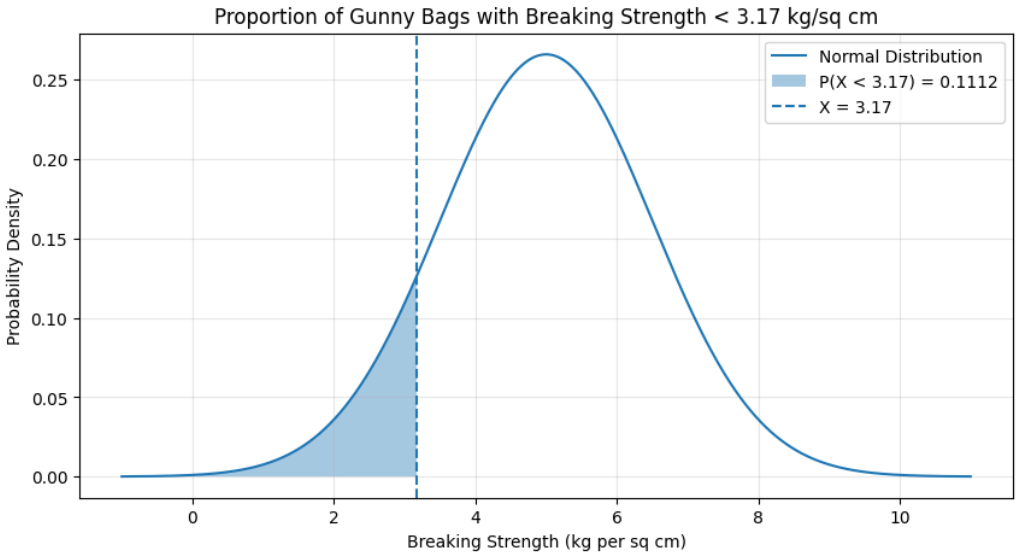


Figure 1: Shaded area for P(X < 3.17)

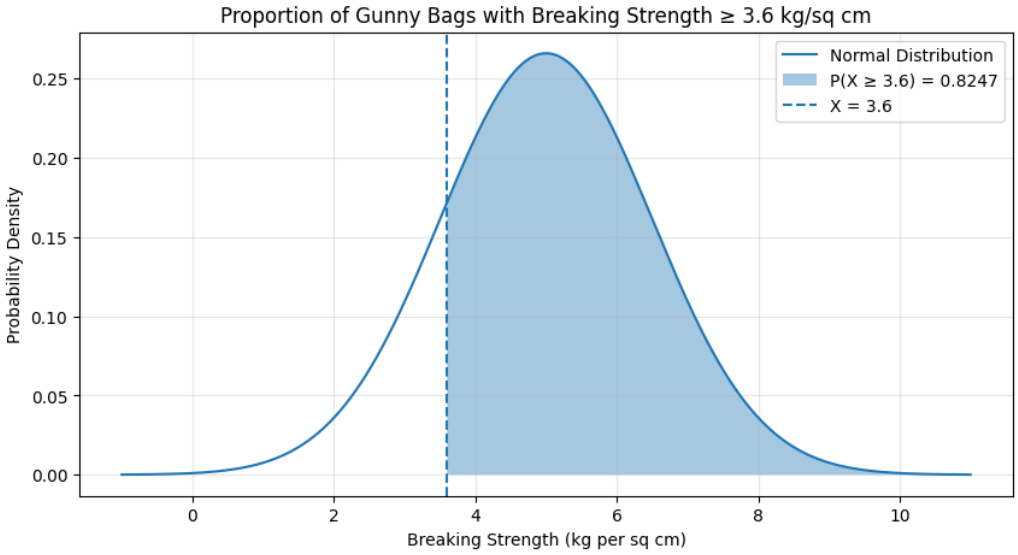


Figure 2: Shaded area for P(X >= 3.6)

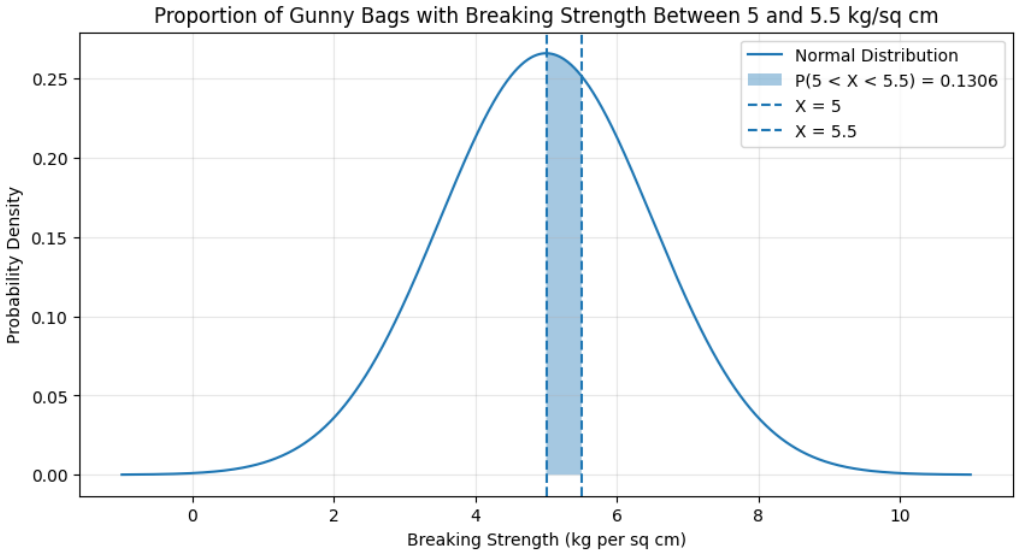


Figure 3: Shaded area for P(5 < X < 5.5)

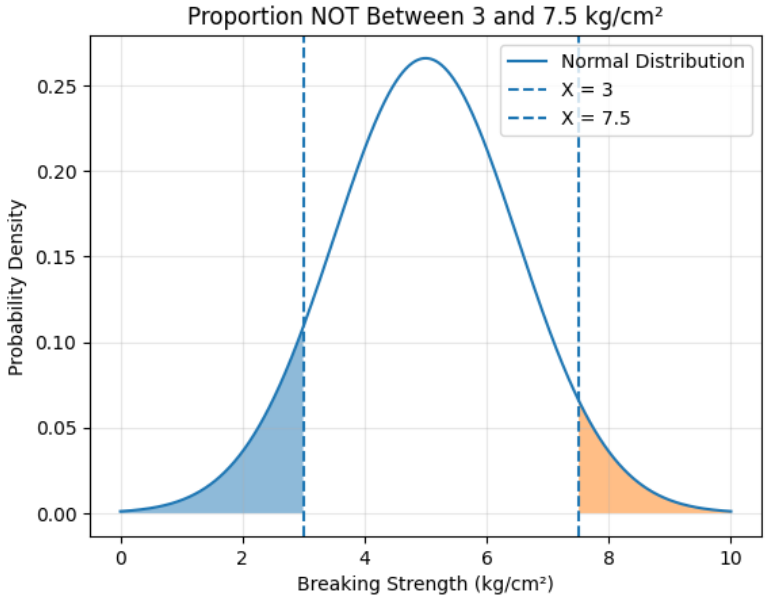


Figure 4: Shaded tails for P(X not between 3 and 7.5)

Odd things and zooming in

The proportion between 5 and 5.5 is small (13%) even though 5 is the mean. This is because the range is only 0.5 wide, which is one third of a standard deviation. Also the answer for "NOT between 3 and 7.5" is found by adding both tails, the left tail (Z = -1.333) and the right tail (Z = 1.667), instead of subtracting.

4. Translating it back to Business Solution

About 11% of the bags break under 3.17 kg per sq cm, which means roughly 1 bag in 9 is weak. On the positive side, about 82% of the bags can hold at least 3.6 kg per sq cm. Around 14% of the bags fall outside the 3 to 7.5 range, so if the company has a quality limit of 3 to 7.5, that is the share of bags that fail it. To improve quality the company should try to reduce the spread (sigma) of its process, because a smaller sigma would push more bags into the safe range.

Problem 3: Hardness of Unpolished Stones (Zingaro Company)

1. Understanding the Business Problem

Zingaro Company makes stones used for printing. For good printing the stones should have a hardness of at least 150. The company thinks the unpolished stones are too soft and not suitable for optimum printing. The question is whether the data supports this belief.

2. Translating it into a Data Science Problem

This is a one sample hypothesis test on a mean with unknown population standard deviation, so a t-test is used. Because the worry is that the stones are too soft, the test is left tailed.

H0: mean hardness of unpolished stones = 150 (or more)
H1: mean hardness of unpolished stones < 150

3. Solving the Data Science Problem

Predict before exploring

The company already thinks the stones are soft, so I expected the sample mean to be below 150 and the test to reject H0.

Analysis

Measure	Value
Sample size	75
Sample mean hardness	134.11
t-statistic	-4.165
Critical value (5%, left tail, df = 74)	-1.666
p-value	0.00004

Odd things and zooming in

Nothing looked odd here. The t-statistic is far below the critical value, and the p-value is much smaller than 0.05, so the result does not depend on the exact choice of significance level.

4. Translating it back to Business Solution

We reject H0. The average hardness of the unpolished stones (134.11) is clearly below 150, so Zingaro is right to believe that the unpolished stones are not good enough for optimum printing. The stones need polishing or some other treatment before they are used for printing.

Problem 4: Hardness of Dental Implants

1. Understanding the Business Problem

The hardness of a metal implant in a dental cavity depends on many things: the method of implant, the temperature at which the metal is treated, the alloy used, and also the dentist, because a dentist may be better at one method than another. The response (hardness) is the variable of interest. A harder implant is better, so the practical question is which method and which dentist combination gives the best implants. The data has 90 rows: 5 dentists, 3 methods, 2 alloys and 3 temperatures (1500, 1600, 1700).

2. Translating it into a Data Science Problem

Hardness is a continuous variable and the factors (dentist, method, alloy) are categories. So the right tools are ANOVA (F-test) to compare means, Levene and Shapiro tests to check the assumptions, and Tukey HSD for the pairwise comparisons. The questions are:

- 4.1 Does the hardness differ between dentists?
- 4.2 Does the hardness differ between methods?
- 4.3 What is the interaction effect between dentist and method for each type of alloy?
- 4.4 How does the hardness vary depending on dentists and methods together?

3. Solving the Data Science Problem

Predict before exploring

I expected the dentists to give roughly similar hardness, and I expected the method to matter more, since the method is the actual procedure. I was not sure about the interaction.

Analyze graphically and numerically

Dentist	Mean	Std dev
1	783.1	129.3
2	786.7	77.4
3	748.6	69.7
4	713.7	178.2
5	676.9	206.5

Method	Mean	Std dev
1	793.9	86.5
2	804.3	95.8
3	627.1	167.6

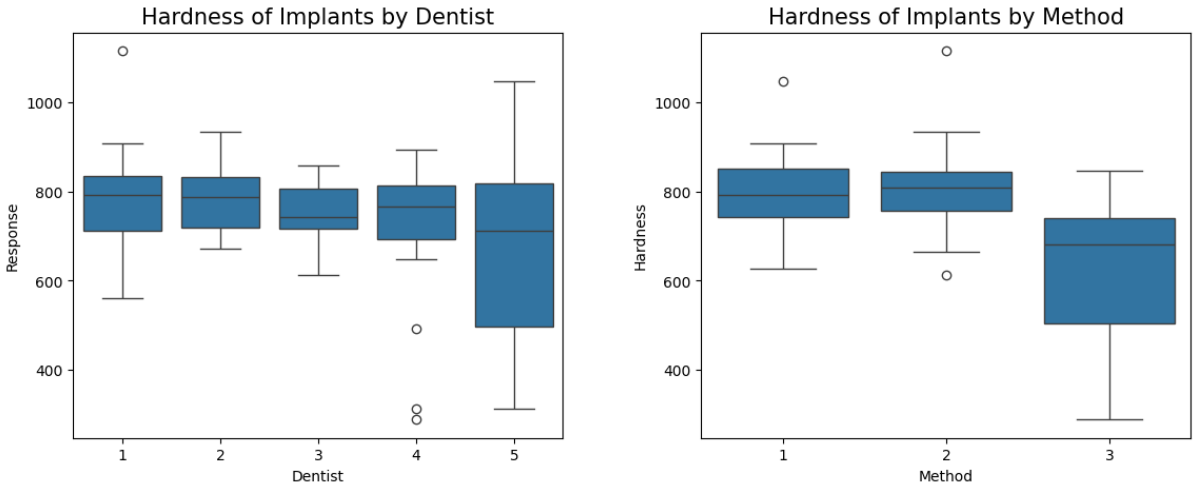


Figure 5: Hardness by Dentist (left) and by Method (right)

4.1 Dentists and 4.2 Methods

The Shapiro test on the full response gave p = 0.000008, so the hardness is not perfectly normal. The sample is balanced (same number of readings in each group), so ANOVA is still fairly safe, but I kept this in mind while reading the results. For dentists, Tukey HSD showed no significant difference between any pair of dentists (all adjusted p-values above 0.15). For methods, the one way ANOVA p-value was 0.00000008, and Tukey showed that Method 3 is significantly lower than both Method 1 (difference about -167) and Method 2 (difference about -177), while Methods 1 and 2 are similar (p = 0.94).

4.3 Interaction of dentist and method for each alloy

Alloy	Dentist (p)	Method (p)	Dentist x Method	Levene (p)
1	0.0115	0.0003	F = 3.40, p = 0.0068	0.313
2	0.3718	< 0.0001	F = 1.92, p = 0.0932	0.783

For Alloy 1 the interaction is significant (p = 0.0068 < 0.05), so the effect of a method depends on the dentist. For Alloy 2 the interaction is not significant at 5% (p = 0.0932), although it is close. Levene's test gave p-values of 0.313 and 0.783, which means equal variances is a fair assumption inside each alloy, so these two ANOVA results can be trusted.

4.4 Dentist and method together (all data)

Source	F	p-value
Dentist	3.55	0.0105
Method	26.70	< 0.0001
Dentist x Method	3.45	0.0020

All three are significant. Tukey on the Dentist x Method groups showed that almost all the significant pairs involve Dentist 5 with Method 3 (mean about 436 for Alloy 1 and 418 for Alloy 2). For Alloy 1, Dentist 4 with Method 3 (mean about 496) also differs from many other groups.

Odd things and zooming in

Two things looked odd. First, the dentist effect is significant in the two way ANOVA (p = 0.0105) but Tukey finds no significant pair of dentists. When I zoomed into the group means, the reason is clear: Dentists 4 and 5 are very low only with Method 3, while with Methods 1 and 2 they are as good as Dentists 1 to 3. So the "dentist effect" is really a dentist and method combination effect. Second, the Levene test fails on the pooled data (p-values 0.0079 for dentist, 0.0041 for method and 0.0039 for dentist x method), which means the spread is not equal across groups. The spread is highest in the Method 3 groups (standard deviation up to about 265), so the pooled p-values should be read with some care.

4. Translating it back to Business Solution

The hardness of dental implants depends mainly on the method used and on the dentist and method combination, not on the dentist alone. Methods 1 and 2 give similar and higher hardness (about 794 and 804). Method 3 gives much softer implants (about 627), and it is especially bad when used by Dentists 4 and 5. Dentists 1 to 3 get stable results with all methods. My recommendation is to use Method 1 or 2 for implants, and to avoid Method 3, especially for Dentists 4 and 5. If Method 3 has to be used, those two dentists may need extra training or review of their procedure.

Overall Conclusion

- **Problem 1:** about 61.7% of the players were injured, and strikers are 31% of the injured players.
- **Problem 2:** about 11% of the gunny bags are weaker than 3.17 kg per sq cm and about 14% fall outside the 3 to 7.5 range.
- **Problem 3:** unpolished stones have a mean hardness of 134.11, which is significantly below 150 ($p = 0.00004$), so they are not suitable for optimum printing.
- **Problem 4:** method has the strongest effect on implant hardness. Method 3 is much weaker than Methods 1 and 2, and the dentist effect comes from Dentists 4 and 5 using Method 3.

Limitations

- In Problem 4 each dentist, method and alloy group has only 3 readings (one per temperature), so the pairwise tests have limited power.
- The pooled hardness data is not normal and the pooled Levene test fails, so the pooled p-values should be treated with some caution.
- The results are only about the data that was given and cannot be generalised to other clinics, alloys or temperatures.